



STEADYHAND

Product Requirements Document

v1.0 · March 2026

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One angle. One shot. Steady wins.

1. Executive Summary

SteadyHand is a daily iOS skill game built around a single mechanic: tilt your phone to a secret angle and hold it perfectly still. Every day, a new target angle is revealed at the moment you tap Start. You get one attempt. Your score is your average deviation in degrees during the hold window. The lower, the better.

The game draws on the same instinct that makes games like Wordle and Globble compelling — a shared daily challenge, one shot, and a leaderboard that resets at midnight. But SteadyHand is physical. It tests body control, not knowledge or reflexes. A surgeon and a coffee addict will score very differently.

The core loop is 30 seconds long. The competitive hook lasts all day.

2. Problem & Opportunity

The Problem

Mobile gaming is dominated by either long-session games (RPGs, strategy) or mindless tap-loops. There is a gap in the market for games that are:

- Extremely short to play (under 60 seconds per session)
- Physically skilled — not luck-based or reflex-based
- Socially competitive in a low-pressure daily format
- Novel enough to be a conversation starter

The Opportunity

The gyroscope and accelerometer in every modern iPhone are almost entirely unused by consumer apps in creative ways. SteadyHand exploits a sensor that has been sitting idle — turning precision motion control into a competitive sport. The Wordle model proved that a single-attempt daily game with a global leaderboard generates organic sharing and habit formation without monetisation pressure.

3. Target Audience

Primary

- Ages 18–35
- Competitive casual gamers
- Daily Wordle / NYT Games players
- People who share game scores on social media
- iPhone users (iOS 16+)

Secondary

- Surgeons, artists, athletes — any community with a professional interest in hand steadiness
- Friend groups looking for a shared daily challenge
- Streamers / content creators (watchable, reactable format)

4. Core Game Loop

The entire game is a single linear flow. There are no menus to navigate mid-attempt, no retries, and no distractions once the player taps Start.

Step-by-Step Flow

Step	State	Description
1	Home	Player sees today's date, their personal best, and the global leaderboard position. The target angle is hidden — shown as "??°".
2	Pre-Start Gate	Player taps START. App requires 2–3 seconds of continuous phone movement (shake/tilt) to confirm the device is in-hand. A visual motion ring fills as they move.
3	Angle Reveal	The target angle flashes on screen (e.g. "47°") for 1 second before the live tilt UI appears. No time to set up on a surface.
4	Targeting	A live dial shows current tilt. Player tilts toward the target. A tolerance arc ($\pm 2^\circ$) is shown. Timer has not started yet.
5	Lock & Hold	Once within $\pm 2^\circ$ for 0.5 seconds, the hold timer starts (default 10 seconds). Haptic confirms lock. Timer counts down.
6	Drift Penalty	If player drifts outside $\pm 2^\circ$ during the hold, the timer pauses and resets to 10 seconds. Attempt is still valid but score degrades.
7	Completion	Timer reaches zero. Score calculated as mean absolute deviation during the 10-second hold window.
8	Score Screen	Score displayed (e.g. 0.34°). Daily rank shown. Share card generated. Leaderboard updates in real time.

Scoring Formula

Score = mean absolute deviation (MAD) of gyroscope pitch/roll readings sampled at 50Hz during the 10-second hold window, expressed in degrees to 2 decimal places.

Lower score = better. A perfect score of 0.00° is theoretically impossible — this is intentional. Human tremor always produces some noise, and that noise is what the game measures.

5. Anti-Cheat System

The most important design constraint: the game must be impossible to win by placing the phone on a fixed surface. A surface eliminates tremor and allows perfect mechanical accuracy. The following layered system makes surface-cheating detectable and attempt-stacking impossible.

Layer 1 — Shake Gate (Pre-Start)

Before the angle is revealed, the player must hold the phone and move it continuously for 2–3 seconds. The accelerometer must register a minimum motion variance threshold (e.g. $>0.8 \text{ m/s}^2$ RMS) throughout this window. A phone placed on a surface cannot pass this gate without human intervention.

Layer 2 — Angle Reveal Timing

The target angle is revealed only after the shake gate passes. It is displayed for exactly 1 second before the live targeting UI appears. This eliminates any opportunity to read the angle, place the phone, and step away — the timer is already running by the time a surface setup could be completed.

Layer 3 — Micro-Tremor Validation

Human hands held in the air always produce imperceptible micro-tremor: a characteristic noise signature of approximately $0.05\text{--}0.15^\circ$ of oscillation. A phone resting on a hard surface produces near-zero noise ($< 0.01^\circ$). During the hold window, if the tremor signature drops below the human threshold for more than 0.5 consecutive seconds, the attempt is flagged as invalid and not submitted to the leaderboard.

Layer 4 — Single Attempt Lock

Each player receives exactly one attempt per 24-hour cycle, keyed to their authenticated account. Once an attempt is submitted — valid or invalid — no further attempts are permitted until the next angle drops at midnight. This prevents score-farming via repeated attempts.

Layer 5 — Orientation Constraint

The game only accepts attempts with the phone in portrait orientation and within 30° of vertical. Angles that are physically achievable only by resting the phone on a surface (e.g. 5° on a near-flat table) are excluded from the daily angle seed pool.

6. Daily Angle System

The daily angle is the same for every player worldwide. It is deterministically generated from the calendar date using a seeded pseudo-random number generator — no server call is required to derive the angle, but the server validates it server-side to prevent spoofing.

Angle Generation Rules

- Range: 15° to 75° (excludes near-horizontal and near-vertical extremes)
- Axis: alternates between pitch and roll on a weekly schedule to vary the physical challenge
- No repeat within a 30-day rolling window — enforced by the seed algorithm
- Seed formula: $\text{SHA-256}(\text{"steadyhand"} + \text{YYYY-MM-DD}) \bmod \text{valid_angle_range}$

Reveal Schedule

- New angle activates at 00:00 UTC daily
- Players receive a push notification at their local morning time (8:00 AM, configurable)
- Angle is publicly visible on the home screen before the player starts — they know what they're aiming for, but cannot pre-aim without passing the shake gate

7. Leaderboard Architecture

SteadyHand uses a three-tier competitive structure. Daily competition drives daily engagement. The all-time leaderboard provides long-term status and retention.

Tier	Scope	Resets	Prize
Daily	Global — all players	Midnight (local server)	   medals
Weekly	Rolling 7-day medal count	Every Monday 00:00 UTC	Weekly badge
All-Time	Cumulative gold → silver → bronze	Never	Hall of Fame tier

End-of-Day Medal Ceremony

At midnight UTC, the daily leaderboard freezes. A push notification is sent to all players who submitted an attempt that day, announcing the top 3. The notification is personalised: ranked players receive their medal; unranked players see where they finished.

A shareable static card is generated for each medal winner showing their score, rank, and the day's angle. This is the primary viral sharing mechanism.

Tiebreaking

- Primary: lowest mean absolute deviation (MAD)
- Secondary: earliest submission time (UTC timestamp)
- Tertiary: account creation date (longest-standing member wins)

8. Profile & Progression

Player profiles accumulate medals over time. The profile is the long-term progression layer — it answers the question "what does winning daily actually mean over months?"

Profile Elements

- Display name + avatar (Apple profile photo, optional)
- Medal cabinet: gold / silver / bronze counts, displayed prominently
- Personal best score (all-time lowest MAD)
- Current streak: consecutive days with a valid attempt submitted
- Attempts vs. medals ratio: shows consistency
- Rank tier badge: Bronze / Silver / Gold / Platinum / Diamond, based on total medal count

Rank Tier Thresholds

Tier	Required Medals	Badge
Bronze	1–4 total medals	
Silver	5–14 total medals	
Gold	15–29 total medals	
Platinum	30–49 total medals	
Diamond	50+ total medals	

9. UX & Interface Design

The UI philosophy is tension through simplicity. Every screen removes non-essential elements. The game should feel like a precision instrument, not a casual app.

Screen Inventory

- Home Screen — today's angle (hidden), personal best, leaderboard top 5, START button
- Shake Gate Screen — motion ring animation, instructional copy, progress indicator
- Angle Reveal Screen — large angle number, 1-second flash, no interaction required
- Targeting Screen — live dial, current angle readout, $\pm 2^\circ$ tolerance arc, lock indicator
- Hold Screen — hold timer, deviation live readout, subtle haptic pulses on drift
- Score Screen — final score, daily rank, share card, leaderboard button
- Leaderboard Screen — daily / weekly / all-time tabs, top 100 list
- Profile Screen — medal cabinet, stats, rank tier badge

Haptic Design

- Shake gate: light continuous rumble while moving
- Angle reveal: single medium impact
- Lock acquired: double tap (confirm)
- Drift out of range: single soft pulse
- Hold complete: long strong impact + completion chime
- Medal awarded: triple impact pattern

Visual Language

- Dark mode first — increases perceived precision and focus
- Single accent colour (deep blue #1A56DB) on a near-black background
- The dial is the centrepiece — large, circular, analogue-feeling
- Typography: SF Pro Display for numbers, SF Pro Text for body
- No decorative illustrations — every element is functional

10. Technical Architecture

Layer	Technology	Rationale
Sensors	CoreMotion / DeviceMotion	Gyroscope + accelerometer at up to 100Hz
Frontend	SwiftUI	Native iOS, smooth animations, haptic APIs
Auth	Sign in with Apple	One tap, privacy-first, required for leaderboards
Backend	Supabase (PostgreSQL)	Real-time leaderboard queries, free tier for MVP

Notifications	APNs (Apple Push)	EOD medal ceremony + daily angle drop alerts
Analytics	PostHog (self-host)	Attempt rates, churn, daily active players

Sensor Specification

- Framework: CoreMotion (CMMotionManager)
- Update rate: 100Hz during shake gate; 50Hz during targeting and hold
- Data used: CMDeviceMotion.attitude (pitch + roll in radians, converted to degrees)
- Tremor detection: standard deviation of 0.1-second rolling window, sampled at 50Hz
- Surface detection threshold: tremor SD < 0.01° for > 0.5s triggers invalid flag

Backend Schema (Supabase)

- users — id, display_name, apple_sub, created_at, streak, total_attempts
- daily_angles — date (PK), angle_degrees, axis (pitch|roll)
- attempts — id, user_id, date, score_mad, submitted_at, valid, tremor_flag
- medals — id, user_id, date, medal_type (gold|silver|bronze)
- leaderboard_daily — materialised view, refreshed on insert to attempts

11. Notifications

Trigger	Message	Timing
Daily angle drop	"Today's angle is live. One shot. Good luck."	8:00 AM (local, configurable)
EOD — medal winner	"You placed 🏆 today with 0.34°. Check your medal."	00:01 UTC
EOD — non-winner	"Today's winner held 0.34°. Tomorrow's angle drops in Xh."	00:01 UTC
Streak reminder	"You're on a 7-day streak. Don't break it."	6:00 PM if no attempt yet
Streak broken	"Your streak ended. Reset and rebuild."	00:01 UTC if no attempt

12. Monetisation Strategy

The MVP launches free with no ads and no paywalls. The competitive integrity of the leaderboard depends on all players having identical experiences. Monetisation is introduced only after the core competitive loop is validated.

Phase 1 — Free (Launch)

- No ads, no IAP, no premium tier

- Focus: grow daily active users and leaderboard depth

Phase 2 — Cosmetic IAP (Post-launch)

- Custom dial skins (does not affect gameplay)
- Custom medal frame designs for the share card
- Profile badge customisation
- All purely cosmetic — zero competitive advantage

Phase 3 — Optional Subscription

- "SteadyHand Pro" — extended stats, historical angle archive, device comparison
- Private friend leaderboards (group up to 20 people)
- Early access to experimental game modes (dual-axis challenges, etc.)

13. Success Metrics

Metric	Target (90 days post-launch)	Measurement
Daily Active Users	5,000+	PostHog DAU
D7 Retention	> 40%	Cohort analysis
D30 Retention	> 20%	Cohort analysis
Avg session length	< 90 seconds	CoreMotion event timestamps
Attempt rate	> 70% of DAU attempt each day	Supabase attempts table
Share card shares	> 15% of scorers share	UTM-tagged share sheet
Invalid attempt rate	< 3%	tremor_flag column
App Store rating	≥ 4.5 stars	App Store Connect

14. Development Milestones

Phase	Milestone	Timeline
1	Core mechanic: motion detection, angle targeting, timer + scoring	Week 1–2
2	Anti-cheat: tremor detection, shake gate, one-attempt lock	Week 2–3
3	Daily angle seed, local leaderboard, profile + medals	Week 3–5
4	Global leaderboard (Supabase), Sign in with Apple, push notifications	Week 5–7
5	Polish: haptics, animations, EOD medal ceremony, share card	Week 7–8
6	Beta testing, bug fixes, App Store submission	Week 8–10

15. Risks & Mitigations

Risk	Severity	Mitigation
Gyroscope variance differs between iPhone models	Medium	Normalise sensor readings per device model; test on iPhone 12 through 16 Pro
Surface-cheat bypass via vibration motor trick	Low	Multi-layer anti-cheat; flag outlier scores for manual review
Leaderboard feels empty at launch (cold start)	High	Seed with 50–100 beta tester scores before public launch; show % rank not absolute rank when < 100 players
One-attempt rule causes frustration	Medium	Communicate clearly in onboarding; offer a "practice mode" with no leaderboard submission
App Store rejection (sensor usage policy)	Low	CoreMotion is standard API; document usage in privacy manifest; no location or biometric data collected
Daily habit fails to form	High	Streak system + morning push notification; make the 30-second session feel ritualistic, not demanding

16. Out of Scope (v1.0)

The following features are explicitly excluded from the v1.0 release to maintain focus and ship velocity.

- Android version — iOS-only for launch; CoreMotion has no direct Android equivalent
- Multiplayer / real-time head-to-head — asynchronous leaderboard only
- Custom challenge creation — daily angle is global and fixed
- Wearable integration (Apple Watch) — phone-only for MVP
- Dual-axis challenges (pitch AND roll simultaneously) — single axis per day
- Social graph / follow system — anonymous global leaderboard only
- In-app purchases — introduced post-validation only

17. Open Questions

- **Hold duration: 10 seconds is the current default. Should this vary by difficulty tier (easy 5s, standard 10s, hard 15s)?**
 - Proposal: fixed at 10s for v1.0 to keep scoring comparable across the global leaderboard
- **Tolerance window: $\pm 2^\circ$ is current. Is this too easy for most players or too hard for casual users?**
 - Proposal: run closed beta and measure mean scores; adjust before public launch
- **Android roadmap: should the v1.0 scope include an Android spike using SensorManager?**

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- Proposal: defer — CoreMotion sensor fidelity is higher and Android fragmentation complicates anti-cheat
 - **Practice mode: should unlimited practice attempts be allowed with no leaderboard submission?**
 - Proposal: yes — reduces friction for new users; must be clearly separated from competitive mode
 - **Angle visibility: should today's angle be visible before tapping START, or hidden until the shake gate?**
 - Proposal: visible on home screen — players should be able to mentally prepare; anti-cheat gates prevent pre-positioning
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End of Document — SteadyHand PRD v1.0